

# Continuous random variables: probability density functions, cumulative density function, quantiles

Prof. Michael Levine

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# What is a continuous random variable?

- ▶ For a real-valued random variable, a function  $f(x)$  is the probability density function if, for all  $-\infty < a \leq b < \infty$ ,

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

- ▶ In other words, the probability  $P(a \leq X \leq b)$  is an area under the curve  $f(x)$  between  $x = a$  and  $x = b$
- ▶ In particular, to be a density function, we must have  $f(x) \geq 0$  and  $\int_{-\infty}^{\infty} f(x) dx = 1$ .
- ▶ Note that a pdf may not, in general, be bounded from above since it is not a probability  $P(X = x)$ !

# Cumulative density function of $X$

- ▶ Let  $X$  be a continuous random variable with a pdf  $f(x)$
- ▶ The cdf

$$F(x) = P(X \leq x) = P(X < x) = \int_{-\infty}^x f(t) dt$$

- ▶ At any point  $x_0$  at which  $f(x)$  is continuous, the CDF  $F(x)$  is differentiable and  $F'(x_0) = f(x_0)$ .

# Independence

- ▶ Here, we give a definition of independence that is true for *any* random variables.
- ▶ Let  $X_1, X_2, \dots, X_n$  be  $n$  random variables defined on the same sample space  $\Omega$ . We say that  $X_1, X_2, \dots, X_n$  are independent if

$$P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$

for  $\forall x_1, x_2, \dots, x_n$ .

- ▶ An equivalent requirement would be

$$P(X_1 > x_1, X_2 > x_2, \dots, X_n > x_n) = \prod_{i=1}^n P(X_i > x_i)$$

# Density examples

- ▶  $f(x) = 1$ , if  $0 \leq x \leq 1$  and 0 otherwise - a uniform density on  $[0, 1]$
- ▶ Its CDF is  $F(x) = x$  if  $0 \leq x \leq 1$ , 0 for  $x < 0$  and 1 for  $x > 1$
- ▶  $f(x) = 3x^2$ , if  $0 \leq x \leq 1$  and 0 otherwise
- ▶ Its CDF is  $F(x) = x^3$  if  $0 \leq x \leq 1$
- ▶  $f(x) = 6x(1 - x)$ , if  $0 \leq x \leq 1$  and 0 otherwise
- ▶ Verify that it is a pdf by direct integration and find its CDF

# Using the density to calculate a probability

- ▶ Let  $X \sim U[0, 1]$ ; consider events  
 $A = \{X \text{ is between } 0.4 \text{ and } 0.6\}$  and  $B = \{X(1 - X) \leq .21\}$
- ▶ Confirm by direct calculation that  $P(A) = .2$
- ▶ Check that  $x(1 - x) = 0.21$  has two roots in  $[0, 1]$  :  $x = 0.3$   
and  $x = 0.7$ . Moreover,  $x(1 - x) \leq .21$  if  $x \leq .3$  or  $x \geq 0.7$
- ▶ Then,  $P(B) = P(X \leq .3) + P(X \geq 0.7) = 0.6$

# Quantiles

- ▶ For a continuous random variable  $X \sim F(x)$  any  $m : F(m) = 0.5$  is a median of  $X$
- ▶ Note that a median may not be unique if the CDF is not strictly increasing
- ▶ Consider  $F(x) = 1 - e^{-x}$  for  $x \geq 0$  and 0 otherwise
- ▶ This function is clearly a CDF; it is also differentiable at any  $x$  except  $x = 0$
- ▶ Define the pdf  $f(x) = F'(x) = e^{-x}$  at any  $x \neq 0$  and  $f(x) = 1$  when  $x = 0$
- ▶ The resulting density is called a **standard exponential density**
- ▶ If  $F(m) = 0.5$  we have  $m = \log 2 = 0.693$

# Quantiles

- ▶ Let  $X$  have the CDF  $F(x)$ . Let  $0 < p < 1$ . The  $p$ th quantile or  $p$ th percentile of  $X$  is the first  $x$  such that  $F(x) \leq p$ :

$$F^{-1}(p) = \inf\{x : F(x) \geq p\}.$$

- ▶ The function  $F^{-1}(p)$  (sometimes denoted  $Q(p)$ ) is called the **quantile function** of  $F$  or  $X$
- ▶  $Q(.25)$  and  $Q(.75)$  are the first and third **quartiles** of  $F$  or  $X$ .

# Generating New Distributions from the Old I

- ▶ Let  $f(x)$  be any density function. Then, for any real  $\mu$  and  $\sigma > 0$

$$g(x) = g_{\mu,\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x - \mu}{\sigma}\right)$$

is also a valid density function

- ▶  $g_{\mu,\sigma}(x)$  is called a **location-scale parameter density**

# Generating New Distributions from the Old II

- ▶ Let  $f_1, \dots, f_k$  be densities for some  $k$ ,  $2 \leq k < \infty$
- ▶ Let  $p_1, \dots, p_n$  be constants such that  $p_i \geq 0$  and  $\sum_{i=1}^k p_i = 1$
- ▶ The function

$$f(x) = \sum_{i=1}^k p_i f_i(x)$$

is a valid density function

- ▶  $f(x)$  is called a **mixture density**. A mixture density is a standard way of generating **heavy-tailed** density functions

# Generating New Distributions from the Old III

- ▶ Let  $F_1, \dots, F_k$  be CDFs for some  $k$ ,  $2 \leq k < \infty$
- ▶ Let  $p_1, \dots, p_n$  be constants such that  $p_i \geq 0$  and  $\sum_{i=1}^k p_i = 1$
- ▶ The function

$$F(x) = \sum_{i=1}^k p_i F_i(x)$$

is a valid CDF

# Generating New Distributions from the Old IV

- ▶ Let  $F$  be a CDF and  $\alpha$  any positive real number.
- ▶ Then,

$$G(x) = F^\alpha(x)$$

is also a valid CDF